Placement Preparation Notes

A sample study pack for testing a RAG application. Each topic starts on a new page so that source citations can be checked page by page.

1. Java: Object-Oriented Programming

What is inheritance in Java?

Inheritance is an OOP mechanism in which one class (the subclass or child class) acquires the fields and methods of another class (the superclass or parent class). It is implemented with the **extends** keyword. Inheritance promotes code reuse and models an 'is-a' relationship, for example a Dog is an Animal. Java supports single inheritance of classes, multilevel inheritance and hierarchical inheritance. Java does not support multiple inheritance of classes, to avoid the diamond problem, but a class can implement multiple interfaces.

What is encapsulation?

Encapsulation means wrapping data (fields) and the methods that operate on that data into a single unit, a class, and restricting direct access to the fields. This is achieved by declaring fields private and exposing them through public getter and setter methods. Encapsulation improves security, maintainability and flexibility.

1. Java: Object-Oriented Programming (continued)

What is polymorphism in Java?

Polymorphism means 'many forms'. It allows the same method name or interface to behave differently depending on the object that uses it. Java supports two types. **Compile-time polymorphism** (method overloading) happens when several methods in the same class share a name but have different parameter lists. **Runtime polymorphism** (method overriding) happens when a subclass provides its own implementation of a method already defined in its superclass; the method that runs is decided at runtime based on the actual object type, which is called dynamic method dispatch.

What is abstraction?

Abstraction hides implementation details and shows only the essential features of an object. In Java it is achieved with abstract classes and interfaces. An abstract class can have both abstract and concrete methods, while an interface traditionally declares only method signatures that implementing classes must define.

Difference between an abstract class and an interface

An abstract class can hold instance variables, constructors and concrete methods, and a class can extend only one abstract class. An interface defines a contract, and a class can implement many interfaces. Use an abstract class when subclasses share state and behaviour, and an interface when unrelated classes must provide the same capability.

2. Java: Collections and Exceptions

ArrayList vs LinkedList

ArrayList is backed by a dynamic array. It gives fast random access by index in O(1) time but slow insertion or deletion in the middle, which is O(n). LinkedList is a doubly linked list. It is fast for insertions and deletions when the position is known but slow for random access.

HashMap internals

A HashMap stores key-value pairs in an array of buckets. The key's hashCode decides the bucket. If two keys land in the same bucket, a collision occurs and entries are chained in a linked list, which becomes a balanced tree when the chain grows long. Average time for get and put is O(1). HashMap allows one null key and is not thread-safe.

Checked vs unchecked exceptions

Checked exceptions, such as IOException, are verified by the compiler, and the code must either handle them with try-catch or declare them with throws. Unchecked exceptions, such as NullPointerException and ArrayIndexOutOfBoundsException, extend RuntimeException and are not checked at compile time. The finally block always runs, whether or not an exception occurs.

3. DBMS

What is normalization?

Normalization is the process of organising data in a relational database to reduce redundancy and avoid insertion, update and deletion anomalies. A table is split into smaller related tables connected through keys. **First Normal Form (1NF)** requires atomic values with no repeating groups. **Second Normal Form (2NF)** requires 1NF and no partial dependency of a non-key attribute on part of a composite key. **Third Normal Form (3NF)** requires 2NF and no transitive dependency between non-key attributes. **BCNF** is a stricter version of 3NF in which every determinant must be a candidate key.

What are ACID properties?

ACID describes reliable transactions. **Atomicity**: a transaction happens completely or not at all. **Consistency**: a transaction moves the database from one valid state to another. **Isolation**: concurrent transactions do not interfere with each other. **Durability**: once committed, changes survive failures.

3. DBMS (continued)

Types of keys

A primary key uniquely identifies each row and cannot be null. A candidate key is any column set that could serve as the primary key. A foreign key refers to the primary key of another table and enforces referential integrity. A composite key is made of two or more columns. A unique key enforces uniqueness but allows a null value.

SQL joins

INNER JOIN returns only rows that match in both tables. LEFT JOIN returns all rows from the left table and matching rows from the right, with NULL where there is no match. RIGHT JOIN is the mirror image. FULL OUTER JOIN returns all rows from both tables. A self join joins a table with itself.

DELETE vs TRUNCATE vs DROP

DELETE removes selected rows, can use a WHERE clause and can be rolled back. TRUNCATE removes all rows quickly and keeps the table structure. DROP removes the table itself, including its structure.

4. Operating Systems

What is the difference between a process and a thread?

A process is an independent program in execution with its own memory space, resources and process control block. A thread is the smallest unit of execution inside a process. Threads of the same process share the code, data and heap, but each thread has its own stack and registers. Creating and switching between threads is cheaper than for processes, and threads communicate easily through shared memory. A crash in one process does not directly affect another, whereas a faulty thread can bring down its whole process.

What is a deadlock?

A deadlock is a situation where a set of processes are blocked because each holds a resource and waits for a resource held by another. Four conditions must hold together: mutual exclusion, hold and wait, no preemption and circular wait. Deadlocks can be prevented by breaking one of the conditions, avoided with the Banker's algorithm, or detected and recovered from.

CPU scheduling algorithms

First Come First Served runs processes in arrival order. Shortest Job First picks the shortest burst time and minimises average waiting time but can starve long jobs. Round Robin gives each process a fixed time quantum and suits time-sharing systems. Priority scheduling runs the highest priority first, and aging prevents starvation.

5. Python

What is the difference between a list and a tuple in Python?

A list is mutable, so its elements can be changed, added or removed after creation, and it is written with square brackets. A tuple is immutable, so it cannot be changed once created, and it is written with parentheses. Tuples are slightly faster, use less memory, and can be used as dictionary keys because they are hashable. Lists are used for collections that change, and tuples for fixed records.

Explain recursion with an example

Recursion is a technique where a function calls itself to solve a smaller version of the same problem. Every recursive function needs a base case that stops the recursion and a recursive case that moves toward the base case. Example: the factorial of n is n multiplied by the factorial of n minus 1, with the base case that factorial of 0 is 1. Without a base case the function recurses forever and raises a RecursionError.

What are decorators and generators?

A decorator is a function that wraps another function to add behaviour without changing its code, applied with the @ syntax. A generator is a function that uses the yield keyword to produce values one at a time, which saves memory for large sequences.

6. Aptitude

Important topics in quantitative aptitude

The most frequently tested quantitative aptitude topics in campus placements are percentages, profit and loss, simple and compound interest, ratio and proportion, time and work, time speed and distance, averages, mixtures and alligations, permutations and combinations, probability, number systems, and ages. Practise formulas daily and solve timed sets to build speed.

Logical reasoning and verbal ability

Logical reasoning covers blood relations, seating arrangements, coding-decoding, syllogisms, direction sense, number and letter series, and puzzles. Verbal ability covers reading comprehension, sentence correction, synonyms and antonyms, para jumbles and fill in the blanks. Data interpretation uses tables, bar graphs, pie charts and line graphs, where quick approximation saves time.

7. HR Interview

What are common questions asked during an HR interview?

Common HR questions include: Tell me about yourself. What are your strengths and weaknesses? Why should we hire you? Why do you want to join our company? Where do you see yourself in five years? Are you willing to relocate or work in shifts? Describe a time you handled conflict in a team. What is your greatest achievement?

How to answer 'Tell me about yourself'

Keep the answer to about one minute. Start with your name and education, mention two or three relevant skills or projects, include one achievement, and end with why you are interested in the role. Do not recite your resume line by line, and do not share unrelated personal details.

Strengths and weaknesses

Choose strengths that match the job and back each with a short example. For weaknesses, name a real but non-critical one and explain what you are doing to improve it.

8. Resume and Projects

What points should be included when explaining a final-year project?

When explaining a final-year project, cover these points: the problem statement and why it matters, the technologies and tools used, the system architecture or workflow, your specific role and contribution, the biggest challenge you faced and how you solved it, the results or outcome, and what you would improve in the future. Be ready for follow-up questions on every technology you list.

Resume preparation tips

Keep the resume to one page for freshers. Include education, skills, projects, internships and achievements. Use action verbs and measurable results. Make sure every skill listed can be defended in an interview. Use a clean format with consistent fonts and no spelling mistakes.